

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer



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Artificial Intelligence for Science and Technology

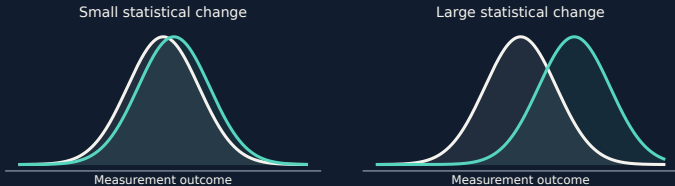
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In collaboration with the **Italian Space Agency (ASI)**

Project: *Quantum Computers for Space Exploration*

Fisher information and parameter estimation

Local sensitivity quantifies the distinguishability of nearby parameter values.



Fisher information

How sensitive the observed statistics are to a small parameter change.

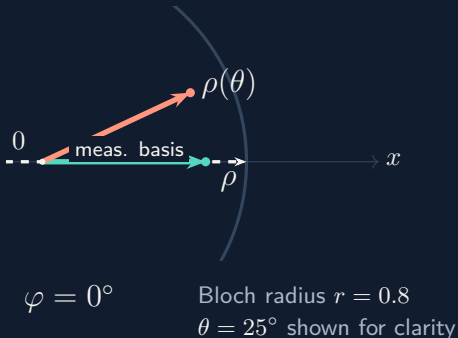
Quantum Fisher information (QFI) is the best local information available from the state, optimized over measurements.

Illustrative distributions. QFI is a local benchmark; attainable precision also depends on the measurement and estimation assumptions.

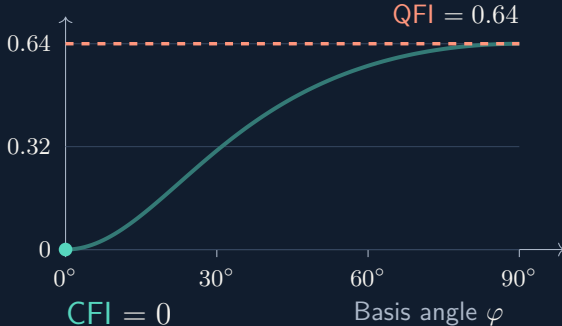
Measurement direction and accessible information

Two states inside the Bloch sphere; the dashed measurement axis rotates.

Two mixed states



Accessible information



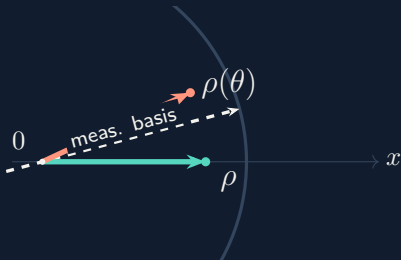
The measurement changes CFI; the QFI of the state family remains fixed.

Mixed states, $r = 0.8$; finite separation shown for clarity. CFI is local at $\theta = 0$ (radians), not a two-state finite difference. Appendix J.

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Two states inside the Bloch sphere; the dashed measurement axis rotates.

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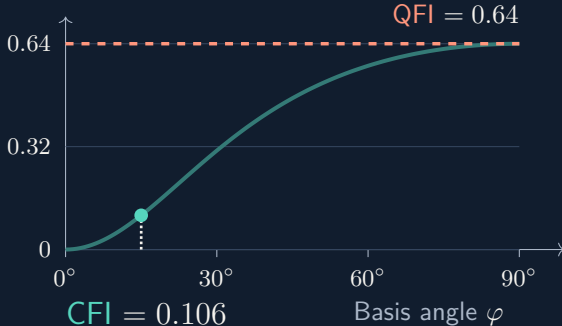


$$\varphi = 15^\circ$$

Bloch radius $r = 0.8$

$\theta = 25^\circ$ shown for clarity

Accessible information



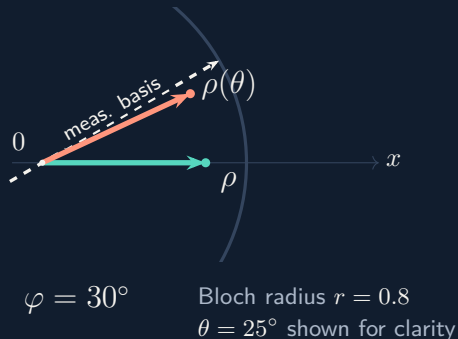
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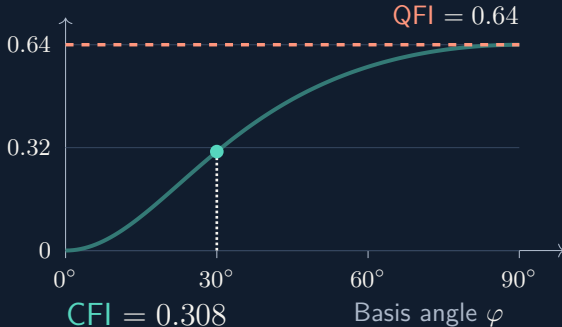
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Two mixed states



Accessible information



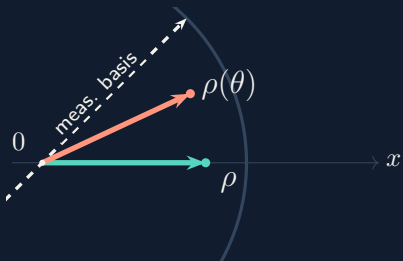
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Measurement direction and accessible information

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Two mixed states

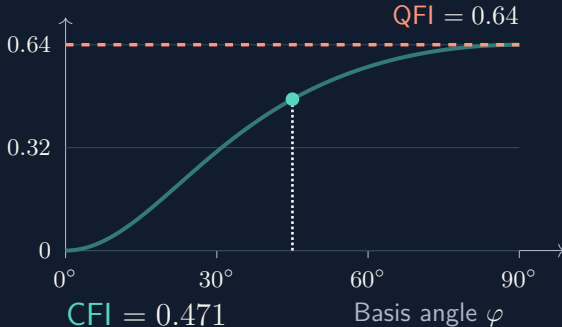


$$\varphi = 45^\circ$$

Bloch radius $r = 0.8$

$\theta = 25^\circ$ shown for clarity

Accessible information



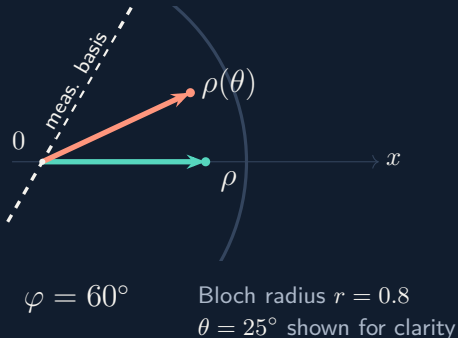
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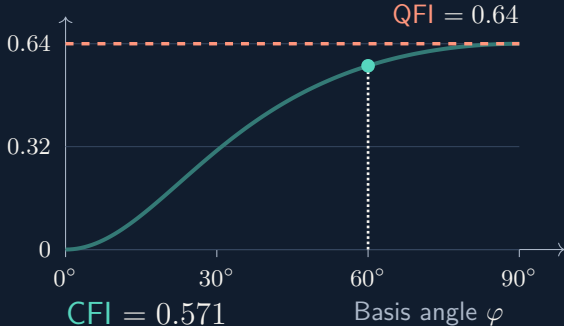
Measurement direction and accessible information

Two states inside the Bloch sphere; the dashed measurement axis rotates.

Two mixed states



Accessible information



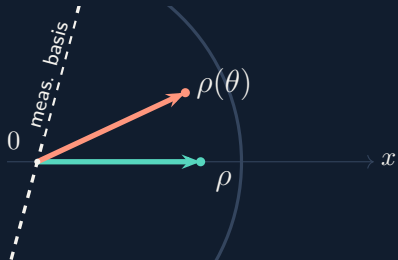
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Measurement direction and accessible information

Two states inside the Bloch sphere; the dashed measurement axis rotates.

Two mixed states

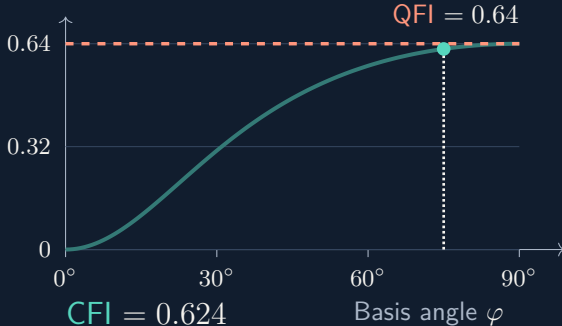


$$\varphi = 75^\circ$$

Bloch radius $r = 0.8$

$\theta = 25^\circ$ shown for clarity

Accessible information



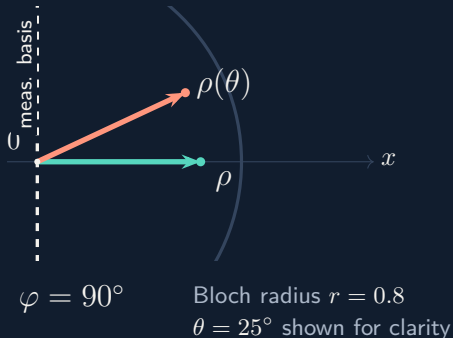
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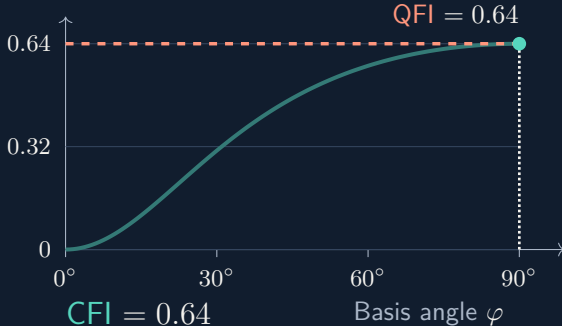
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VQSE and principal-component extraction

Variational Quantum State Eigensolver: dominant spectral components of a mixed state.

PCA / COVARIANCE MATRIX

$$C = \sum_i s_i \mathbf{v}_i \mathbf{v}_i^T$$

Dominant directions capture the largest share of variance.

VQSE / DENSITY MATRIX

$$\rho = \sum_i \lambda_i |v_i\rangle \langle v_i|$$

Dominant state components carry the largest probability weights.

$$V \rho V^\dagger \approx \text{diag}(\lambda_1, \lambda_2, \dots)$$

Train a basis rotation V .
Frequencies estimate eigenvalues;
 $V^\dagger|i\rangle$ recovers eigenvectors.

The retained eigenspace provides a compact input to fidelity-based QFI bounds.

Implemented and benchmarked established VQSE / VQFIE methods: Cerezo et al. (2022); Beckey et al. (2022). No speedup claim.

Local metrology in the transverse-field Ising model

Six interacting spins encode a magnetic field; measurements access a four-spin subsystem.



Periodic spin chain

4 retained / 2 discarded

Ground state Lowest-energy preparation

Depolarized Fixed depolarizing noise

Thermal Field-dependent populations

Compare preparation families over the operating field range.

Transverse-field Ising model (TFIM), $N = 6$, $n = 4$, periodic boundaries, $J = 1$. Thesis: NoiseAnalysis, Project 9.

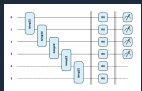
Parallel numerical and quantum-circuit workflows

The same accessible state connects exact calculations to variational spectral estimation.

$$H(h_x) = -J \sum_j Z_j Z_{j+1} - h_x \sum_j X_j, \quad U_{h_x}(t) = e^{-itH(h_x)}$$

Preparation / encoding

$$\begin{aligned}\rho_G &= |\psi_0\rangle\langle\psi_0| \\ \rho_\beta &= e^{-\beta H} / Z_\beta \\ \rho(t) &= U \rho_{\text{in}} U^\dagger\end{aligned}$$



Prepare
and encode
with $\tilde{U}_{h_x}$

Access

$$\rho_S = \text{Tr}_E \rho$$

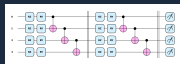
Partial trace
over E

Retain S ;
leave E
unmeasured.

Spectrum

$$\rho_S = \sum_i \lambda_i |v_i\rangle\langle v_i|$$

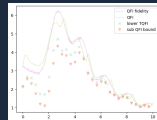
Exact spectrum
(PCA analogy)



VQSE basis rotation

Information

Exact SLD QFI
TQFI bounds

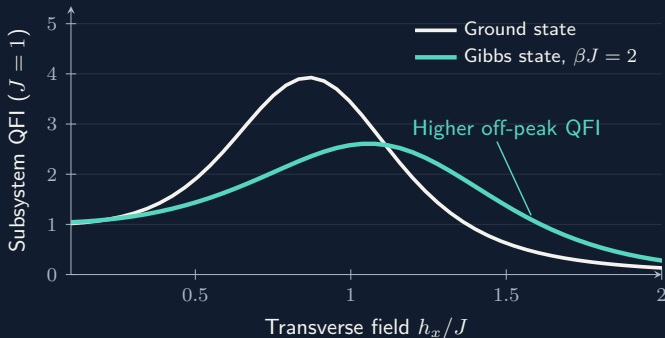


Interchange implementations at matched interfaces while preserving the physical order.

Numerical operators above; schematic circuits below. $\tilde{U}$ approximates U with gates. Gibbs preparation is a separate process.

Thermal sensitivity beyond the ground-state optimum

A lower peak can coexist with greater sensitivity away from the optimum.



At $\beta J = 2$

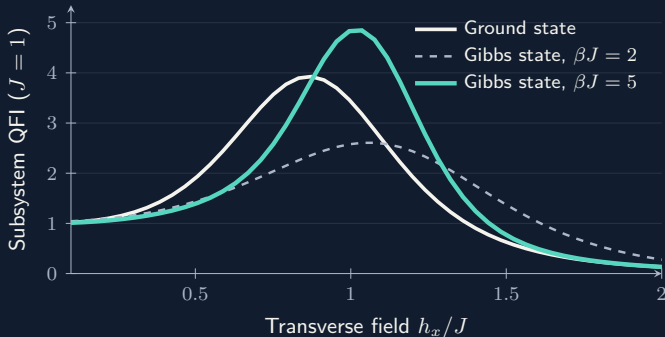
Peak QFI:
thermal 2.61
ground state 3.93
Higher sensitivity
away from the
ground-state peak.

The preferred preparation depends on the expected field range as well as peak sensitivity.

Exact SLD calculation; periodic $N = 6$, $n = 4$, $J = 1$. Different preparation families; no universal thermal advantage.

Temperature controls the sensitivity profile

Cooling to $\beta J = 5$ produces a higher, more localized peak in this finite system.



+23.5%

higher peak local QFI
at $\beta J = 5$

40/60

sampled fields favour
thermal preparation

Choose temperature for the expected field range: peak sensitivity or a broader response.

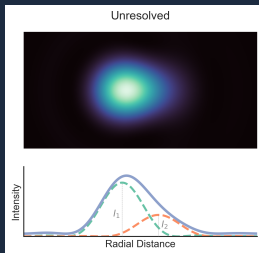
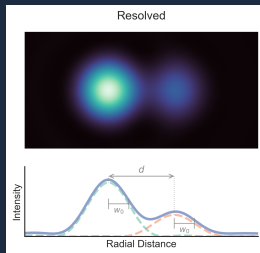
Periodic $N = 6$, $n = 4$, $\beta J = 5$. Metrics: 60 fields on $h_x/J \in [0.1, 2.5]$; plot ends at 2. Exact SLD calculation.

SPADE detector

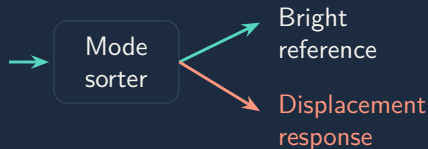
Spatial-mode measurements for exoplanet imaging

A faint companion must be resolved in the presence of a bright reference source.

Direct imaging



Spatial-mode sorting



Photon counts in a spatial-mode basis.

SPADE changes the measurement basis to reveal weak spatial signals.

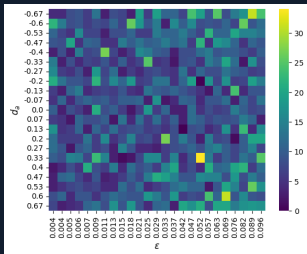
Thesis illustrations; schematic mode sorter. SPADE: spatial-mode demultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

Experimental observations and identifiability

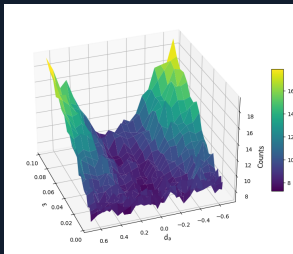
The completed ASI Matera experiment supplied aggregate first-order counts.

52,500 count observations: 525 settings $\times$ 100 repetitions; 10 ms each

ONE REPETITION



MEAN OF 100



Missing observables

Separate modal outputs

Other measurement bases

Incident flux and failure outcomes

Repeated counts constrain the mean response; they do not identify the quantum state.

Thesis: Exoplanet Experiment. d_a : displacement; ϵ : source ratio. Aggregate counts; no basis rotations or experimental TQFI.

Classical probabilities and quantum states

A probability vector has a diagonal quantum representation in a chosen basis.

CLASSICAL PROBABILITY EMBEDDING

$$\mathbf{p} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \longmapsto \rho_{\text{diag}} = \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}$$

Classical channel: $\mathbf{p}' = R\mathbf{p}$

Redistributes probabilities between outcomes.

QUANTUM COHERENCE

$$\rho = \begin{pmatrix} p_0 & c \\ c^* & p_1 \end{pmatrix}$$

Quantum channel: $\rho' = \mathcal{E}(\rho)$

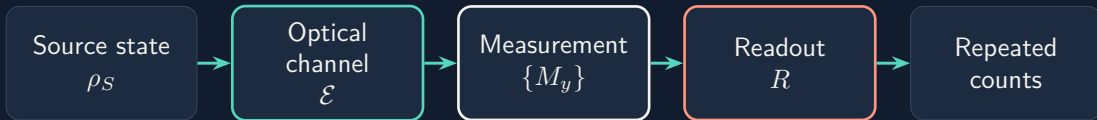
Acts on populations and coherences.

The embedding preserves measured probabilities; coherences require further data.

Two-outcome illustration: $p_i \geq 0$, $p_0 + p_1 = 1$, $|c|^2 \leq p_0 p_1$. A state vector of probabilities does not determine a quantum channel.

A quantum-to-classical channel model

A physical forward model for propagation, measurement and detector reporting.



$$p_y = \text{Tr}[M_y \mathcal{E}(\rho_S)],$$

$$\mathbf{p}_{\text{rep}} = R\mathbf{p}$$

Quantum stages
Loss, dephasing and coherent mixing.

Classical stages
Readout, calibration and count noise.

The framework preserves physical constraints and locates losses of information.

Thesis: sequential channel model. Coherence and optical-channel parameters are not reconstructed from aggregate counts.

Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

Random phase flips

$$\widetilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \widetilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z \rho Z}{2} = \text{diag}(\rho)$$

Kraus freedom changes the representation. Physical hypotheses concern the channel.

Two-mode illustration of the thesis HG-dephasing model at $v = 0$. Equivalent representations cannot be distinguished by system-only data.

A physical construction: mode-dependent loss

Mode transmission $0 \leq \tau_j \leq 1$; loss is represented by an orthogonal failure state.

Choose operators with explicit physical roles

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

Channel validity can be proved mathematically; physical suitability requires calibration.

Derived from the thesis mode-dependent loss map. Input: one-photon mode space after common survival; failure is retained in the output.

Information retained by the optical receiver

Compare quantum and classical information at successive stages of the receiver.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

Checkpoint

Fraction of source QFI

● Resolved reference, first-order and failure outputs	99.68–100%
● Binary readout; assumed 0.35% port confusion	0.13–57.0%

Aggregation and readout confusion can strongly reduce the accessible information.

Conditional model: same nine settings, per photon surviving common loss. Ranges across settings; not reconstructed experimental QFI.

From state information to experimental access

Sensitivity depends on preparation, accessible degrees of freedom and measurement.

- **Variational estimation**

Implemented spectral methods for mixed-state QFI bounds.

Exact numerical calculations benchmark the variational estimates.

- **Preparation over a field range**

Thermal preparation offers higher off-peak sensitivity in the studied regime.

- **A physically constrained observation model**

Physical channels connect information limits to data and guide calibration.

Next: resolved SPADE outputs, independent calibration and a programmable modal basis.

The optical variational protocol VQ-SPADE remains a proposal for a future experiment; QFI geometry for optimization is in Appendix B.

Spectral truncation and the local-QFI limit

Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

F is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cutoff.

At regular operating points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds need not be pointwise bounds on the exact SLD QFI.

Thesis: Methods; Mathematical Background. Cerezo et al. (2022); Beckey et al. (2022).

The ansatz metric and its scope

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re}[\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz: $F^{\text{ans}} = 4g_{\text{FS}}$, the Fubini–Study metric up to convention.

$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

Stationarity gives the regularized natural-gradient update.

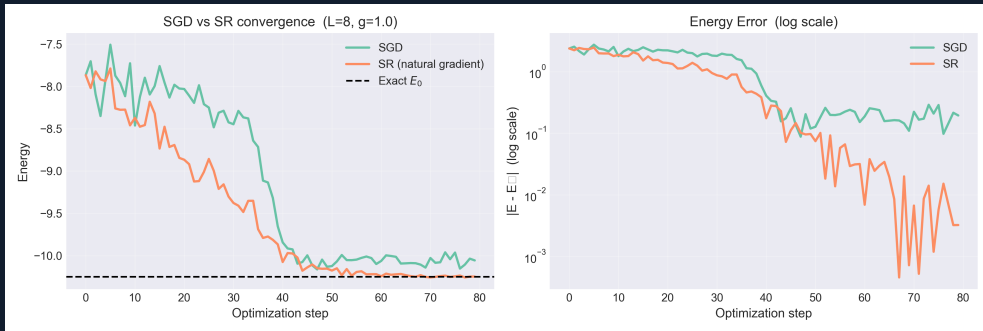
The quadratic term constrains state change, not objective curvature.

Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.

Thesis: Mathematical Background; VQSE. A full metric has quadratic parameter-count overhead; no general training speedup is claimed.

Stochastic reconfiguration in a neural quantum state

A separate pure-state RBM example illustrates metric-based optimization.



SR lowers the energy error in this run; it does not benchmark mixed-state VQSE.

Thesis figure: VQSE chapter. TFIM at $\hbar_x/J = 1$; 80 steps, 300 Monte Carlo samples/step, $\eta = 0.05$, SR $\lambda = 10^{-4}$; same seed.

Thermal sensitivity and data processing

The Gibbs and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

The comparison changes the preparation family; it does not reverse a fixed noise channel.

Thesis: Noise Analysis. Finite system; exact SLD QFI.

Proof of Kraus-unitary freedom

A change of Kraus basis leaves the channel unchanged.

$$\widetilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc}$$

Orthonormal columns of U .

$$\sum_a \widetilde{K}_a \rho \widetilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

A physical realization adds assumptions about the environment; the reduced map alone does not select them.

Standard Kraus equivalence: an isometry, or a unitary after padding with zero operators. Algebra shown explicitly for this presentation.

A classical response does not fix a quantum channel

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic R , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji} \rho_{ii} |j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For $R = I$, both the identity channel and complete dephasing reproduce the same populations.

Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.

Thesis: classical/quantum embedding and HG coherence discussion. Channel-extension example derived for the presentation.

The loss map and its physical checks

The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks: $\tau_j = 1$ transmits mode j ; $\tau_j = 0$ sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the τ_j .

Renormalizing the transmitted branch alone changes the question to postselected performance.

Thesis: mode-dependent loss and flagged mode reduction. The record lacks independent modal-transmission calibration.

Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \xi]$$

Grouped five-fold validation; all components refitted within each training fold.

Held-out structure	RMSE	Standardized SD	95% coverage
Checkerboard cells	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.

Thesis: Exoplanet Experiment Appendix, hierarchical calibration validation. RMSE in counts per 10 ms.

From SPADE to a variational optical measurement

VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

The present record motivates the acquisition design; it does not implement this protocol.

Thesis: proposed VQ-SPADE scheme. Additional independent channel and background calibrations are required.

Mixed states and measurement-dependent information

Two finite-separated vectors illustrate one state family; Fisher information is evaluated locally.

$$\rho(\theta) = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8, \quad \rho \equiv \rho(0)$$

$$M_{\pm}^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_{\pm}(\theta) = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At $\varphi = 0^\circ$: CFI = 0. At 45° : CFI = $8/17 \simeq 0.471$.

At $\varphi = 90^\circ$, measuring Z attains QFI = 0.64; the SLD is $L_0 = rZ$.

Both displayed vectors have radius 0.8; their 25° separation is only illustrative.

Changing the measurement changes accessible information, not the underlying quantum limit.

Exact mixed-qubit model; θ in radians. Full rank keeps every displayed measurement regular. No finite-difference QFI approximation is used.